

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 12A

ENTROPY (SYSTEM, SURROUNDINGS & TOTAL ENTROPY)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 12: ENTROPY AND ENERGETICS
SUB-TOPIC:	12A: Entropy (System, Surroundings & Total Entropy)
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	201 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. Definition of System Entropy****[3 Marks]**
WCH14/01/Jan23/Q5

Entropy is a measure of the degree of disorder of a system.

- (a) Define entropy, S , at a molecular level. [1]
(b) Predict, with a reason, whether entropy increases or decreases for: $\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{g})$. [2]

Q2. 2. Standard System Entropy Calculation**[4 Marks]**
WCH14/01/Oct22/Q6

Calculate ΔS_{system} for: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$
Standard entropies S° ($\text{J K}^{-1} \text{mol}^{-1}$): $\text{N}_2(\text{g})=191.6$, $\text{H}_2(\text{g})=130.6$, $\text{NH}_3(\text{g})=192.3$

- (a) Calculate ΔS_{system} in $\text{J K}^{-1} \text{mol}^{-1}$. [3]
(b) Explain the sign of ΔS_{system} in terms of number of gas moles. [1]

Q3. 3. Entropy Calculation & Feasibility Step 3**[3 Marks]**
WCH14/01/Sample/Q3

For reaction 3: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
(b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

Q4. 4. Entropy Calculation & Feasibility Step 4**[3 Marks]**
WCH14/01/Sample/Q4

For reaction 4: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
(b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

Q5. 5. Entropy Calculation & Feasibility Step 5**[3 Marks]**
WCH14/01/Sample/Q5

For reaction 5: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
(b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

Q6. 6. Entropy Calculation & Feasibility Step 6**[3 Marks]**
WCH14/01/Sample/Q6

For reaction 6: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
(b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

Q7. 7. Entropy Calculation & Feasibility Step 7**[3 Marks]**
WCH14/01/Sample/Q7

For reaction 7: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
(b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

Q8. 8. Entropy Calculation & Feasibility Step 8**[3 Marks]**
WCH14/01/Sample/Q8

For reaction 8: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]

- (b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

[3 Marks]

Q9. 9. Entropy Calculation & Feasibility Step 9

WCH14/01/Sample/Q9

For reaction 9: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
(b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

[3 Marks]

Q10. 10. Entropy Calculation & Feasibility Step 10

WCH14/01/Sample/Q10

For reaction 10: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
(b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

[3 Marks]

Q11. 11. Entropy Calculation & Feasibility Step 11

WCH14/01/Sample/Q11

For reaction 11: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
(b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

[3 Marks]

Q12. 12. Entropy Calculation & Feasibility Step 12

WCH14/01/Sample/Q12

For reaction 12: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
(b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

[3 Marks]

Q13. 13. Entropy Calculation & Feasibility Step 13

WCH14/01/Sample/Q13

For reaction 13: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
(b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

[3 Marks]

Q14. 14. Entropy Calculation & Feasibility Step 14

WCH14/01/Sample/Q14

For reaction 14: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
(b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

[3 Marks]

Q15. 15. Entropy Calculation & Feasibility Step 15

WCH14/01/Sample/Q15

For reaction 15: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
(b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

[3 Marks]

Q16. 16. Entropy Calculation & Feasibility Step 16

WCH14/01/Sample/Q16

For reaction 16: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
(b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

[3 Marks]**Q17. 17. Entropy Calculation & Feasibility Step 17****WCH14/01/Sample/Q17**

For reaction 17: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
- (b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

[3 Marks]**Q18. 18. Entropy Calculation & Feasibility Step 18****WCH14/01/Sample/Q18**

For reaction 18: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
- (b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

[3 Marks]**Q19. 19. Entropy Calculation & Feasibility Step 19****WCH14/01/Sample/Q19**

For reaction 19: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
- (b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

[3 Marks]**Q20. 20. Entropy Calculation & Feasibility Step 20****WCH14/01/Sample/Q20**

For reaction 20: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
- (b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

[3 Marks]**Q21. 21. Entropy Calculation & Feasibility Step 21****WCH14/01/Sample/Q21**

For reaction 21: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
- (b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

[3 Marks]**Q22. 22. Entropy Calculation & Feasibility Step 22****WCH14/01/Sample/Q22**

For reaction 22: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
- (b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

[3 Marks]**Q23. 23. Entropy Calculation & Feasibility Step 23****WCH14/01/Sample/Q23**

For reaction 23: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
- (b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

[3 Marks]**Q24. 24. Entropy Calculation & Feasibility Step 24****WCH14/01/Sample/Q24**

For reaction 24: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
- (b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

Q25. 25. Entropy Calculation & Feasibility Step 25**[3 Marks]****WCH14/01/Sample/Q25**

For reaction 25: $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]
- (b) Calculate ΔS_{total} and state whether the reaction is spontaneous at 298 K. [1]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. A* Challenge: Temperature of Feasibility & Equilibrium K 26****[5 Marks]**
WCH14/01/Hard/Q26

For the thermal decomposition of calcium carbonate:

 $\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q27. 27. A* Challenge: Temperature of Feasibility & Equilibrium K 27**[5 Marks]**
WCH14/01/Hard/Q27

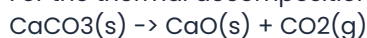
For the thermal decomposition of calcium carbonate:

 $\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q28. 28. A* Challenge: Temperature of Feasibility & Equilibrium K 28**[5 Marks]**
WCH14/01/Hard/Q28

For the thermal decomposition of calcium carbonate:

 $\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q29. 29. A* Challenge: Temperature of Feasibility & Equilibrium K 29**[5 Marks]**
WCH14/01/Hard/Q29

For the thermal decomposition of calcium carbonate:

 $\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q30. 30. A* Challenge: Temperature of Feasibility & Equilibrium K 30**[5 Marks]**
WCH14/01/Hard/Q30

For the thermal decomposition of calcium carbonate:

 $\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q31. 31. A* Challenge: Temperature of Feasibility & Equilibrium K 31**[5 Marks]**
WCH14/01/Hard/Q31

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]

(b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q32. 32. A* Challenge: Temperature of Feasibility & Equilibrium K 32

[5 Marks]
WCH14/01/Hard/Q32

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]

(b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q33. 33. A* Challenge: Temperature of Feasibility & Equilibrium K 33

[5 Marks]
WCH14/01/Hard/Q33

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]

(b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q34. 34. A* Challenge: Temperature of Feasibility & Equilibrium K 34

[5 Marks]
WCH14/01/Hard/Q34

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]

(b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q35. 35. A* Challenge: Temperature of Feasibility & Equilibrium K 35

[5 Marks]
WCH14/01/Hard/Q35

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]

(b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q36. 36. A* Challenge: Temperature of Feasibility & Equilibrium K 36

[5 Marks]
WCH14/01/Hard/Q36

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]

- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q37. 37. A* Challenge: Temperature of Feasibility & Equilibrium K 37

[5 Marks]
WCH14/01/Hard/Q37

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q38. 38. A* Challenge: Temperature of Feasibility & Equilibrium K 38

[5 Marks]
WCH14/01/Hard/Q38

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q39. 39. A* Challenge: Temperature of Feasibility & Equilibrium K 39

[5 Marks]
WCH14/01/Hard/Q39

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q40. 40. A* Challenge: Temperature of Feasibility & Equilibrium K 40

[5 Marks]
WCH14/01/Hard/Q40

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q41. 41. A* Challenge: Temperature of Feasibility & Equilibrium K 41

[5 Marks]
WCH14/01/Hard/Q41

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q42. 42. A* Challenge: Temperature of Feasibility & Equilibrium K 42

[5 Marks]
WCH14/01/Hard/Q42

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q43. 43. A* Challenge: Temperature of Feasibility & Equilibrium K 43

[5 Marks]
WCH14/01/Hard/Q43

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q44. 44. A* Challenge: Temperature of Feasibility & Equilibrium K 44

[5 Marks]
WCH14/01/Hard/Q44

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q45. 45. A* Challenge: Temperature of Feasibility & Equilibrium K 45

[5 Marks]
WCH14/01/Hard/Q45

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q46. 46. A* Challenge: Temperature of Feasibility & Equilibrium K 46

[5 Marks]
WCH14/01/Hard/Q46

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q47. 47. A* Challenge: Temperature of Feasibility & Equilibrium K 47

[5 Marks]
WCH14/01/Hard/Q47

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]

- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q48. 48. A* Challenge: Temperature of Feasibility & Equilibrium K 48

[5 Marks]
WCH14/01/Hard/Q48

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q49. 49. A* Challenge: Temperature of Feasibility & Equilibrium K 49

[5 Marks]
WCH14/01/Hard/Q49

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

Q50. 50. A* Challenge: Temperature of Feasibility & Equilibrium K 50

[5 Marks]
WCH14/01/Hard/Q50

For the thermal decomposition of calcium carbonate:



$\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the minimum temperature (T_{min}) in Kelvin at which this reaction becomes spontaneous ($\Delta S_{\text{total}} \geq 0$). [3]
- (b) Calculate ΔS_{total} at 1200 K and deduce the equilibrium constant K using $\Delta S_{\text{total}} = R \ln K$. [2]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 12A

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Definition of System Entropy 1. (a) Measure of the number of ways of arranging energy/particles in a system (disorder) (1). 1. (b) Increases (1). Gas molecules have higher kinetic energy and far more random arrangements than liquid (1).	[3]
Q2	Q2: 2. Standard System Entropy Calculation 2. (a) $\sum S^\circ(\text{products}) = 2 \times 192.3 = 384.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). $\sum S^\circ(\text{reactants}) = 191.6 + 3(130.6) = 583.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). $\Delta S_{\text{sys}} = 384.6 - 583.4 = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 2. (b) Negative because 4 moles of gas react to form 2 moles of gas $\rightarrow$ decrease in disorder (1).	[4]
Q3	Q3: 3. Entropy Calculation & Feasibility Step 3 3. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 3. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q4	Q4: 4. Entropy Calculation & Feasibility Step 4 4. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 4. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q5	Q5: 5. Entropy Calculation & Feasibility Step 5 5. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 5. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q6	Q6: 6. Entropy Calculation & Feasibility Step 6 6. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 6. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q7	Q7: 7. Entropy Calculation & Feasibility Step 7 7. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 7. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q8	Q8: 8. Entropy Calculation & Feasibility Step 8 8. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 8. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q9	Q9: 9. Entropy Calculation & Feasibility Step 9 9. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 9. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q10	Q10: 10. Entropy Calculation & Feasibility Step 10 10. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 10. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q11	Q11: 11. Entropy Calculation & Feasibility Step 11 11. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 11. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q12	Q12: 12. Entropy Calculation & Feasibility Step 12 12. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 12. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]

Q13	Q13: 13. Entropy Calculation & Feasibility Step 13 13. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 13. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q14	Q14: 14. Entropy Calculation & Feasibility Step 14 14. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 14. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q15	Q15: 15. Entropy Calculation & Feasibility Step 15 15. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 15. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q16	Q16: 16. Entropy Calculation & Feasibility Step 16 16. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 16. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q17	Q17: 17. Entropy Calculation & Feasibility Step 17 17. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 17. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q18	Q18: 18. Entropy Calculation & Feasibility Step 18 18. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 18. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q19	Q19: 19. Entropy Calculation & Feasibility Step 19 19. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 19. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q20	Q20: 20. Entropy Calculation & Feasibility Step 20 20. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 20. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q21	Q21: 21. Entropy Calculation & Feasibility Step 21 21. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 21. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q22	Q22: 22. Entropy Calculation & Feasibility Step 22 22. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 22. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q23	Q23: 23. Entropy Calculation & Feasibility Step 23 23. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 23. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q24	Q24: 24. Entropy Calculation & Feasibility Step 24 24. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 24. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]
Q25	Q25: 25. Entropy Calculation & Feasibility Step 25 25. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 25. (b) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Spontaneous because $\Delta S_{\text{total}} > 0$.	[3]

[illegible]

[illegible]

Q48	Q48: 48. A* Challenge: Temperature of Feasibility & Equilibrium K 48 48. (a) At feasibility boundary $\Delta S_{\text{total}} = 0 \Rightarrow \Delta S_{\text{surr}} = -\Delta S_{\text{sys}} = -161 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). $-\Delta H / T = -161 \Rightarrow T = 178000 / 161 = 1105.6 \text{ K}$ (2). 48. (b) At 1200 K, $\Delta S_{\text{surr}} = -178000/1200 = -148.3$. $\Delta S_{\text{tot}} = 161 - 148.3 = +12.7 \text{ J K}^{-1} \text{ mol}^{-1}$. $\ln K = 12.7 / 8.31 = 1.528 \Rightarrow K = e^{1.528} = 4.61$ (2).	[5]
Q49	Q49: 49. A* Challenge: Temperature of Feasibility & Equilibrium K 49 49. (a) At feasibility boundary $\Delta S_{\text{total}} = 0 \Rightarrow \Delta S_{\text{surr}} = -\Delta S_{\text{sys}} = -161 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). $-\Delta H / T = -161 \Rightarrow T = 178000 / 161 = 1105.6 \text{ K}$ (2). 49. (b) At 1200 K, $\Delta S_{\text{surr}} = -178000/1200 = -148.3$. $\Delta S_{\text{tot}} = 161 - 148.3 = +12.7 \text{ J K}^{-1} \text{ mol}^{-1}$. $\ln K = 12.7 / 8.31 = 1.528 \Rightarrow K = e^{1.528} = 4.61$ (2).	[5]
Q50	Q50: 50. A* Challenge: Temperature of Feasibility & Equilibrium K 50 50. (a) At feasibility boundary $\Delta S_{\text{total}} = 0 \Rightarrow \Delta S_{\text{surr}} = -\Delta S_{\text{sys}} = -161 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). $-\Delta H / T = -161 \Rightarrow T = 178000 / 161 = 1105.6 \text{ K}$ (2). 50. (b) At 1200 K, $\Delta S_{\text{surr}} = -178000/1200 = -148.3$. $\Delta S_{\text{tot}} = 161 - 148.3 = +12.7 \text{ J K}^{-1} \text{ mol}^{-1}$. $\ln K = 12.7 / 8.31 = 1.528 \Rightarrow K = e^{1.528} = 4.61$ (2).	[5]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 12A

FAQ #1: Units of System Entropy vs Enthalpy

Category: Units Conversion

COMMON EXAMINER TRAP / MISTAKE:

Adding ΔS_{system} ($\text{J K}^{-1} \text{mol}^{-1}$) directly to ΔH (kJ mol^{-1}).

MODEL ANSWER / EXAMINER EXPECTATION:

Always convert ΔH to Joules ($\times 1000$) before calculating $\Delta S_{\text{surroundings}} = -\Delta H / T$.
System and surroundings entropy MUST both be in $\text{J K}^{-1} \text{mol}^{-1}$.

FAQ #2: Minus Sign in Surroundings Entropy Formula

Category: Formula Trap

COMMON EXAMINER TRAP / MISTAKE:

Forgetting the minus sign in $\Delta S_{\text{surroundings}} = -\Delta H / T$.

MODEL ANSWER / EXAMINER EXPECTATION:

Exothermic reactions (ΔH negative) lose heat to surroundings $\rightarrow$ surroundings entropy INCREASES (ΔS_{surr} positive). The minus sign ensures this.

FAQ #3: Spontaneity Criterion

Category: Feasibility

COMMON EXAMINER TRAP / MISTAKE:

Stating $\Delta S_{\text{system}} > 0$ is required for spontaneity.

MODEL ANSWER / EXAMINER EXPECTATION:

Spontaneity depends ONLY on TOTAL entropy: $\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} > 0$. An endothermic reaction with negative ΔS_{sys} can still be spontaneous if surroundings entropy is highly positive.

FAQ #4: Temperature of Feasibility Calculation

Category: Threshold Calculation

COMMON EXAMINER TRAP / MISTAKE:

Using $\Delta S_{\text{surroundings}} = \Delta S_{\text{system}}$ at boundary.

MODEL ANSWER / EXAMINER EXPECTATION:

At the threshold of feasibility, $\Delta S_{\text{total}} = 0$, so $\Delta S_{\text{surroundings}} = -\Delta S_{\text{system}}$.
Therefore $-\Delta H / T_{\text{min}} = -\Delta S_{\text{sys}} \Rightarrow T_{\text{min}} = \Delta H / \Delta S_{\text{sys}}$.

FAQ #5: Entropy at Absolute Zero

Category: Third Law

COMMON EXAMINER TRAP / MISTAKE:

Assuming non-zero entropy at 0 K for perfect crystals.

MODEL ANSWER / EXAMINER EXPECTATION:

Third Law of Thermodynamics: The entropy of a pure, perfectly crystalline substance is EXACTLY ZERO at absolute zero (0 K).

FAQ #6: State Changes and Entropy

Category: Qualitative Entropy

COMMON EXAMINER TRAP / MISTAKE:

Predicting entropy decrease during melting or boiling.

MODEL ANSWER / EXAMINER EXPECTATION:

Melting (solid → liquid) and boiling (liquid → gas) ALWAYS increase system entropy due to increased molecular motion and disorder.

FAQ #7: Gas Moles and System Entropy Sign**Category: Qualitative Rules****COMMON EXAMINER TRAP / MISTAKE:**

Ignoring change in number of gas moles when predicting sign of ΔS_{system} .

MODEL ANSWER / EXAMINER EXPECTATION:

If gas moles INCREASE (reactants → products), ΔS_{system} is POSITIVE. If gas moles DECREASE, ΔS_{system} is NEGATIVE.

FAQ #8: Dissolution of Salts and Entropy**Category: Solution Entropy****COMMON EXAMINER TRAP / MISTAKE:**

Assuming dissolving an ionic lattice always increases entropy.

MODEL ANSWER / EXAMINER EXPECTATION:

Dissolving involves lattice breakdown (+S) AND ion hydration (−S due to water ordering). If hydration ordering dominates, ΔS_{system} can be negative.

FAQ #9: Kinetic Inertness vs Thermodynamic Feasibility**Category: Kinetics vs Thermodynamics****COMMON EXAMINER TRAP / MISTAKE:**

Assuming $\Delta S_{\text{total}} > 0$ guarantees a rapid reaction.

MODEL ANSWER / EXAMINER EXPECTATION:

$\Delta S_{\text{total}} > 0$ proves a reaction is THERMODYNAMICALLY FEASIBLE, but high activation energy (E_a) can make it kinetically inert (infinitely slow at room temp).

FAQ #10: Relating ΔS_{total} to Equilibrium Constant K**Category: Thermodynamic Equilibrium****COMMON EXAMINER TRAP / MISTAKE:**

Confusing R value or units in $\Delta S_{\text{total}} = R \ln K$.

MODEL ANSWER / EXAMINER EXPECTATION:

$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$. ΔS_{total} MUST be in $\text{J K}^{-1} \text{ mol}^{-1}$. $\ln K = \Delta S_{\text{total}} / R$. Exponentiate: $K = e^{(\Delta S_{\text{total}} / R)}$.